

Fig 1: One collective, changing interface

The collective is identified with the persistent interior, not a frozen set of boundary variables. Interior nodes (1,2,3) (blue, ringed) are fixed throughout; the screening boundary (orange) is re-inferred and changes identity as z sweeps -1 -> +1.

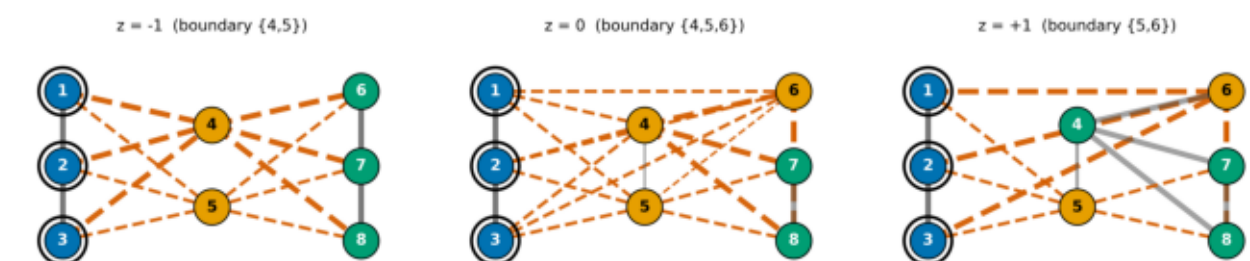


Fig 3: Rate-induced loss of screening

Every frozen configuration along the path has a valid compact ($|B| \leq 3$) exact blanket, yet moving at finite rate transiently and reversibly elevates leakage above 0.

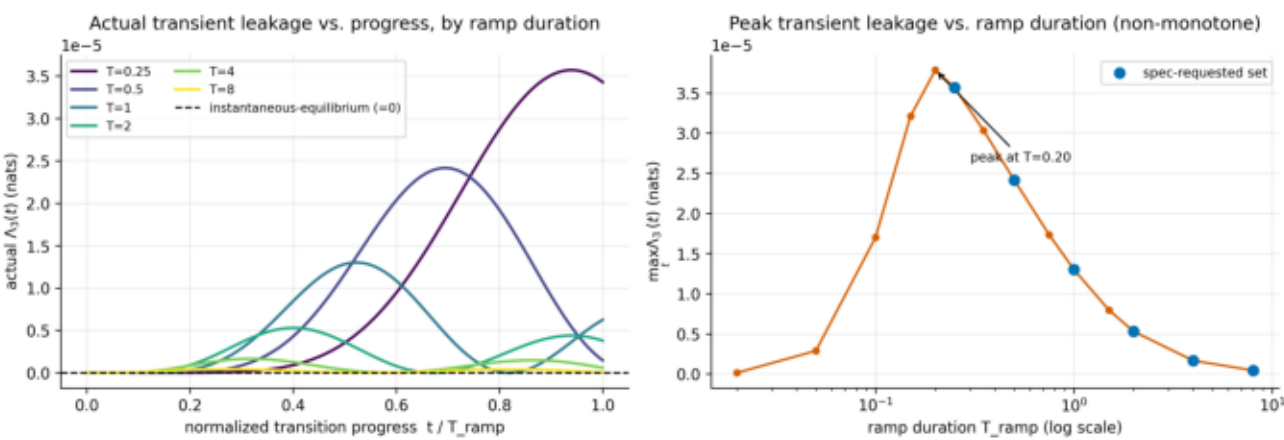


Fig 5: Endpoint validity can hide pathwise failure

Pathwise leakage through control + release, T=2 (coordinated control)

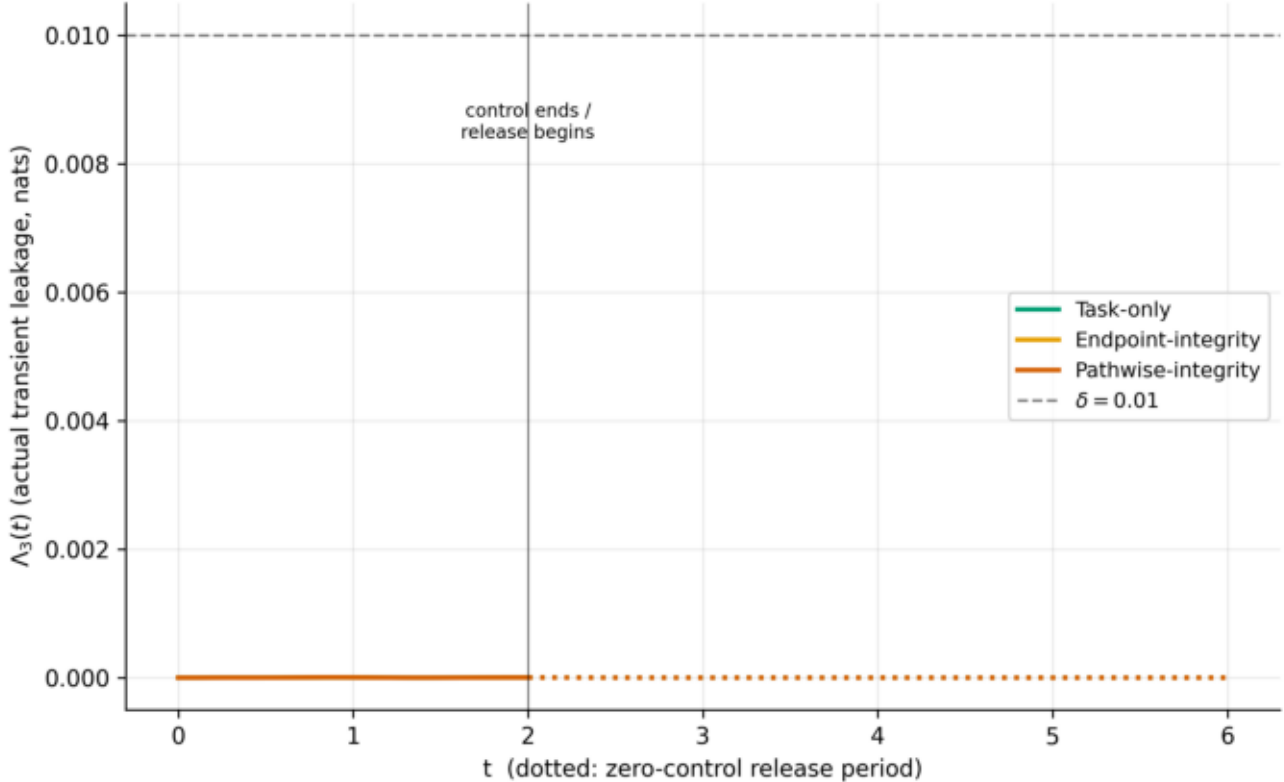


Fig 7: Cost of preserving collective integrity

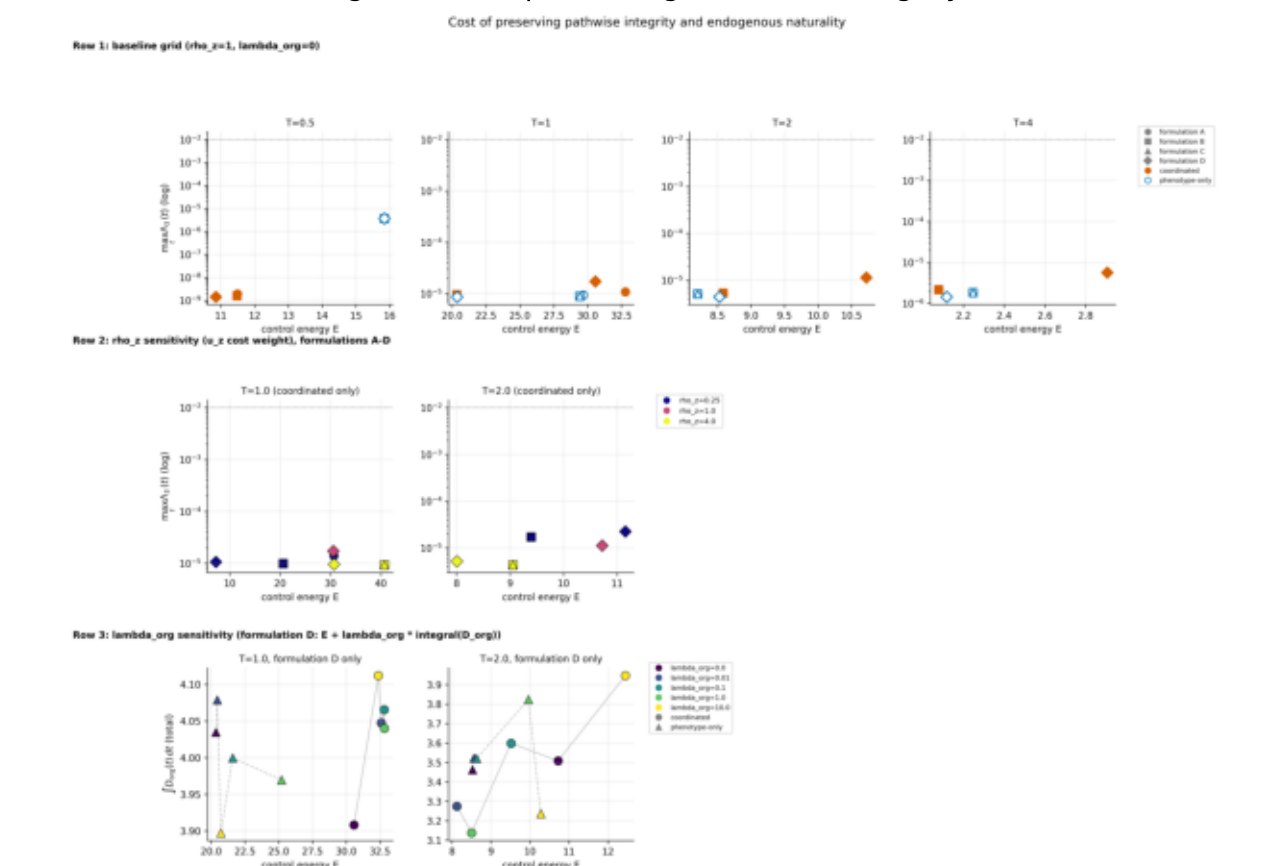


Fig 9: Phenotype-only vs coordinated control

Phenotype-only vs. coordinated control, and sensitivity to the u_x cost weight

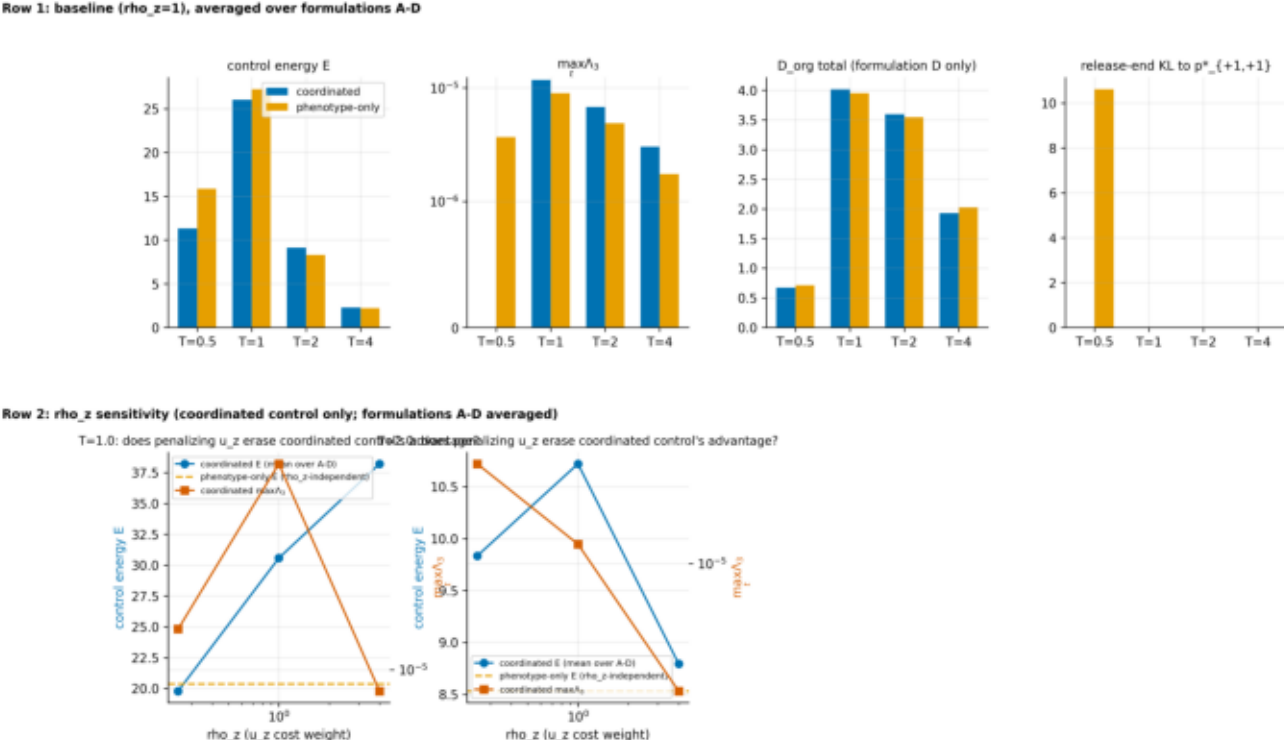


Fig 11: Finite-data integrity classification

Simultaneous-band pathwise classification error vs. sample size (delta_fig11=1e-05, 40 realizations/cell, true classes: T_ramp in [0.1, 0.15, 0.2, 0.25] vs [4, 8])

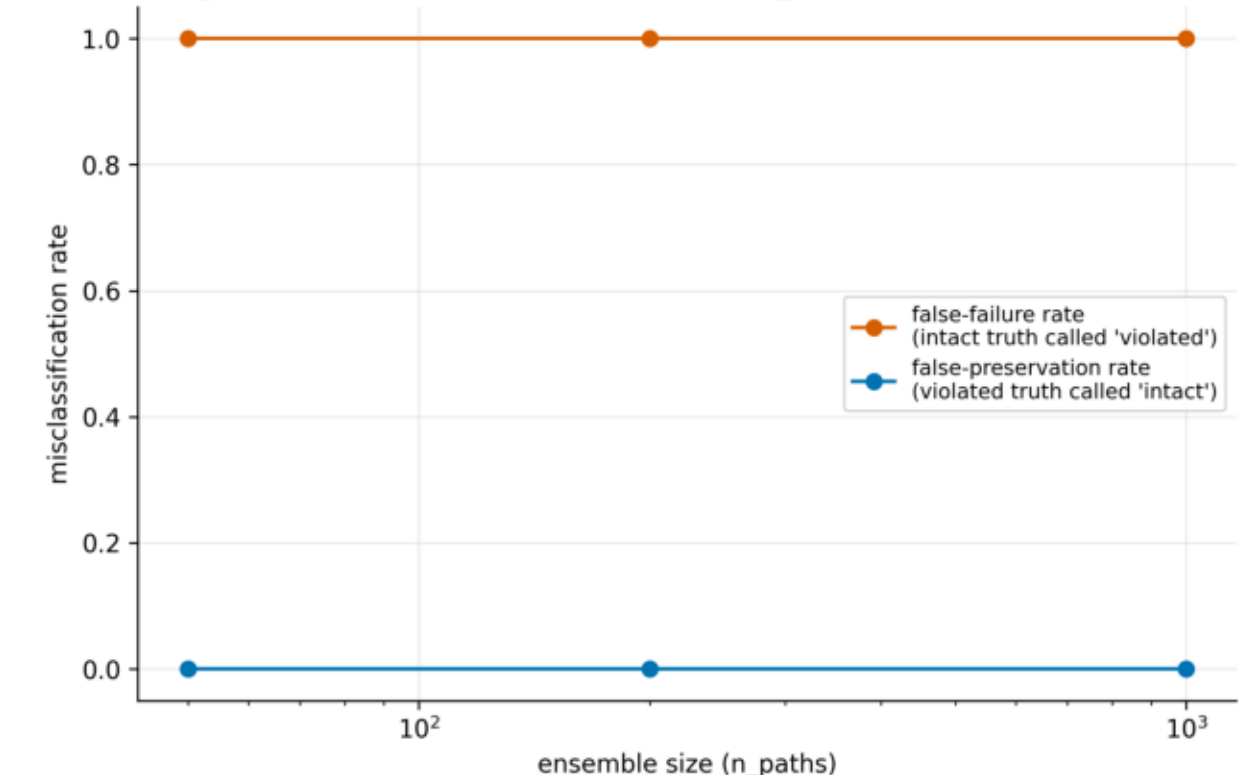


Fig 2: Quasi-static blanket landscape

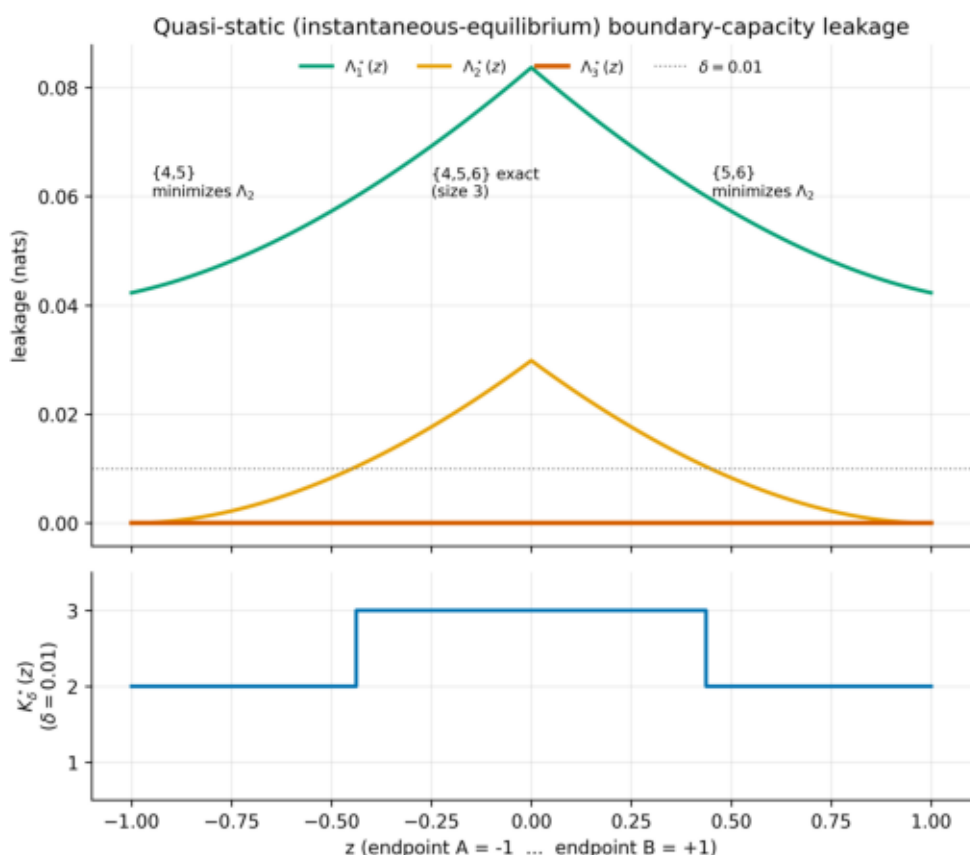


Fig 4: What is being controlled?

Representative protocols at T=2 (coordinated control): task-only vs. pathwise-integrity-constrained -- state trajectories and best-found piecewise-linear controls

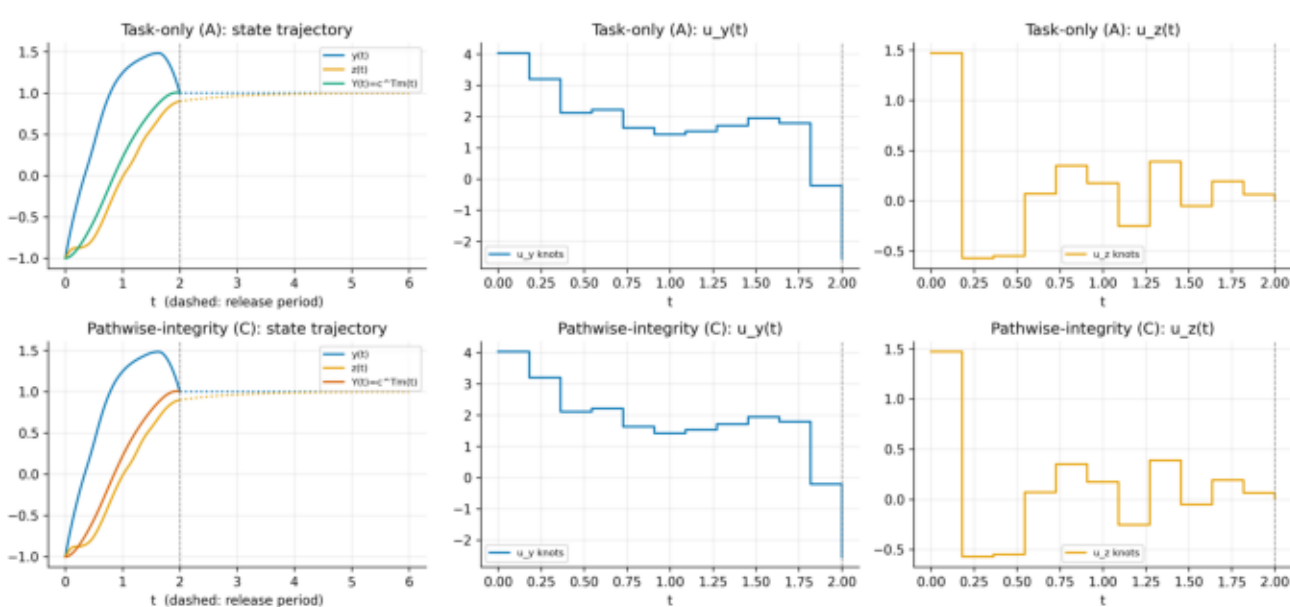


Fig 6: Boundary membership through the transition

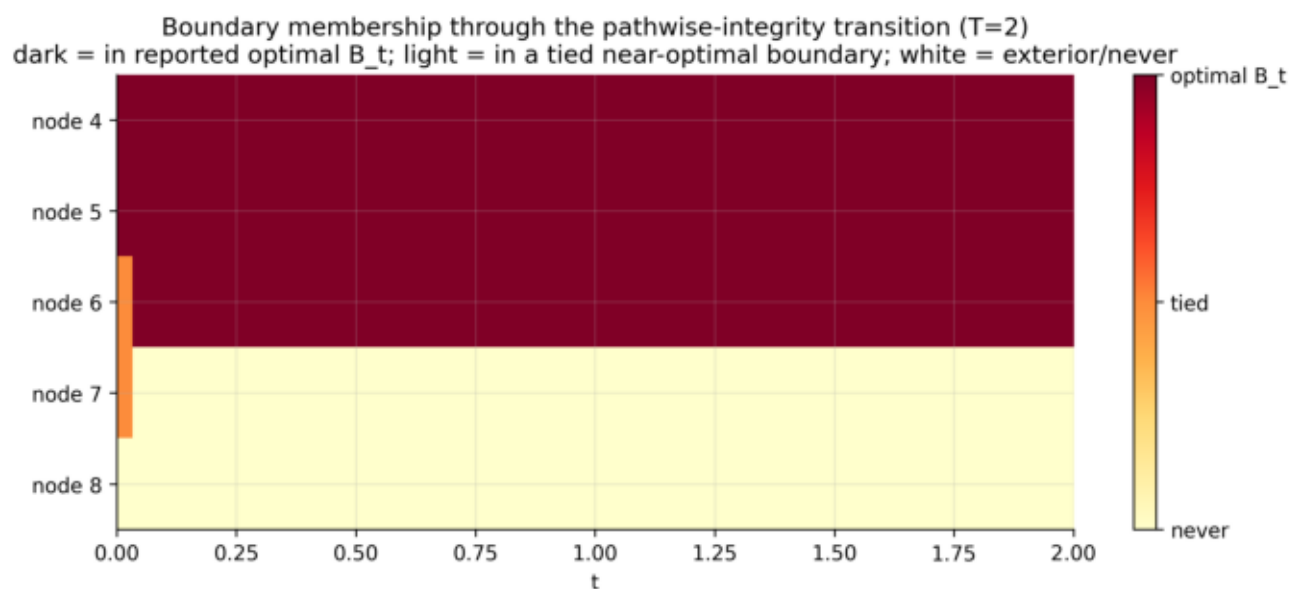


Fig 8: Natural organization vs functional identity

Endogenous-organization departure vs. functional (blanket) leakage, T=2 (both formulations here use lambda_org=0 -- see the lambda_org sweep for the explicit D_org-penalized case)

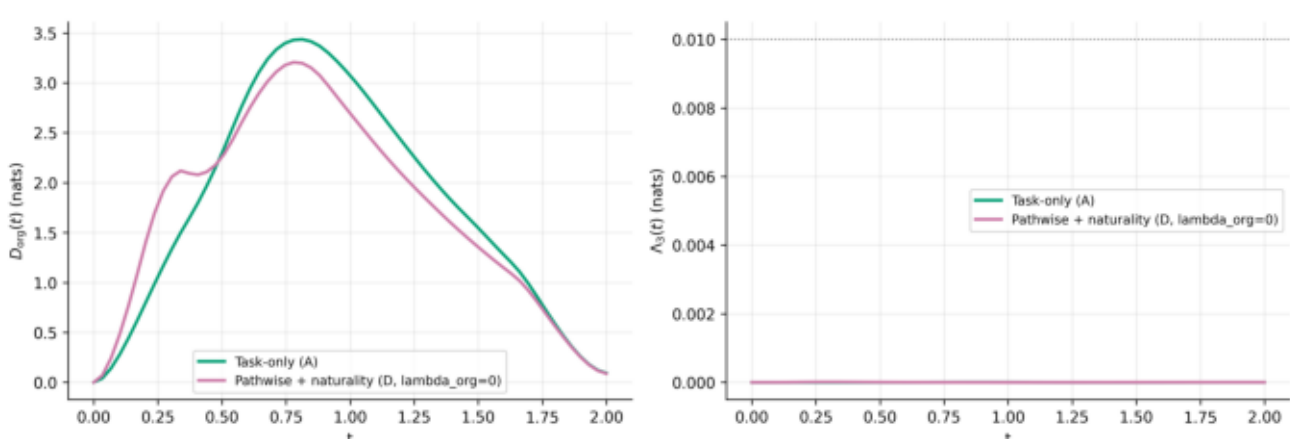


Fig 10: Finite-data recovery

